

House Price Prediction Using Machine Learning

Project Report

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1. Introduction

House price prediction is an important machine learning application in the real-estate domain. Property prices are influenced by several factors, including living area, number of bedrooms and bathrooms, location, condition, waterfront availability, and the age of a property. A data-driven prediction model can identify relationships between these characteristics and property prices.

The main objective of this project is to build a regression-based house price prediction system and compare different machine learning approaches. The project follows a complete machine learning workflow: data inspection, cleaning, exploratory data analysis, visualization, feature preprocessing, train-test splitting, model training, evaluation, and final model selection.

Three regression algorithms were selected for comparison: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. Their performance was measured using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

2. Dataset and Data Preparation

The dataset contains residential property information and uses price as the target variable. The available features include bedrooms, bathrooms, living area, lot size, floors, waterfront, view, condition, above-ground area, basement area, year built, renovation year, city, and state/ZIP code.

The first stage of the project was to inspect the dataset structure. The number of observations, column data types, missing values, duplicate records, and basic price statistics were examined. This inspection helped identify potential data-quality issues before model development.

Records with zero or negative values in the price column were removed because they do not represent valid house prices. The remaining observations were retained for analysis and prediction.

Some columns were excluded from the predictive feature set. The date column was not used for prediction, while street was excluded because it has very high cardinality. Country was also removed because it contains only one value and therefore does not provide meaningful variation for prediction.

After cleaning, price was defined as the target variable and the remaining useful property characteristics were treated as predictor variables.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure of the data and identify relationships between house prices and important property characteristics. Visualization was used as the main method for interpreting the data before applying machine learning models.

Six meaningful visualizations were created as part of the analysis:

1. Distribution of House Price — used to examine the overall distribution and spread of property prices.
2. House Price vs Living Area — used to investigate the relationship between property size and price.
3. Correlation Heatmap — used to identify relationships among numerical variables.
4. House Price vs Number of Bedrooms — used to compare prices across different bedroom counts.
5. House Price vs Number of Bathrooms — used to examine the relationship between bathroom count and price.
6. House Price by Waterfront — used to compare prices of properties with and without waterfront access.

The EDA provided a visual understanding of the dataset and helped identify which property characteristics may have stronger relationships with house prices. These findings also provided useful context for interpreting the machine learning results.

4. Feature Engineering and Preprocessing

Feature preparation was required because the dataset contains both numerical and categorical variables. Numerical features included bedrooms, bathrooms, sqft_living, sqft_lot, floors, waterfront, view, condition, sqft_above, sqft_basement, yr_built, and yr_renovated.

The categorical variables were city and statezip. These variables cannot be directly supplied to the regression algorithms as text values, so One-Hot Encoding was used to convert their categories into numerical indicator features.

A ColumnTransformer was used to apply the appropriate transformation to numerical and categorical columns. The preprocessing was incorporated into the machine learning pipelines so that the same transformation process was consistently applied during model training and prediction.

The cleaned dataset was then divided into training and testing subsets using an 80:20 split with random_state=42. The training data was used to learn the relationships between features and house prices, while the test data was reserved for evaluating model performance on unseen observations.

5. Machine Learning Models

Three regression models were trained according to the project requirements.

Linear Regression was used as the baseline model. It estimates house prices by learning a linear relationship between the predictor variables and the target price.

Decision Tree Regressor was used to capture nonlinear relationships. A decision tree makes a sequence of feature-based decisions and uses the resulting groups to generate price predictions.

Random Forest Regressor was also trained. Random Forest combines predictions from multiple decision trees and can capture more complex relationships between property features and prices.

All three models were trained using the same training and testing framework. This made the resulting metrics directly comparable and allowed the project to identify which model performed best on the test dataset.

6. Model Evaluation

The models were evaluated using four metrics. MAE measures the average absolute difference between actual and predicted prices, where a lower value is better. MSE squares prediction errors and therefore gives greater weight to large errors. RMSE is the square root of MSE and expresses the error in the same units as house price. R² Score measures the proportion of variation in the target that is explained by the model, with a higher value generally indicating better performance.

The test-set results were:

Model	MAE	MSE	RMSE	R ² Score
Linear Regression	139,638.89	5.4014e+10	232,409.25	0.6369
Decision Tree	149,182.95	9.0211e+10	300,352.01	0.3936
Random Forest	121,307.27	1.0233e+11	319,894.63	0.3121

The results show that Random Forest has the lowest MAE, while Linear Regression has the strongest overall results when R² Score, MSE, and RMSE are considered together.

7. Model Comparison and Final Selection

The evaluation results show different strengths among the three models. Random Forest achieved the lowest MAE at 121,307.27, indicating the smallest average absolute error. However, its RMSE was 319,894.63 and its R² Score was 0.3121.

Decision Tree produced an MAE of 149,182.95, an RMSE of 300,352.01, and an R² Score of 0.3936. These results were better than Random Forest for RMSE and R², but still below Linear Regression.

Linear Regression achieved the lowest RMSE at 232,409.25 and the lowest MSE at 5.4014e+10. Most importantly, it achieved the highest R² Score of 0.6369. This means it explained more of the variation in house prices than the other two models in this test.

Although no single metric should be considered in isolation, the combination of R^2 Score, RMSE, and MSE indicates that Linear Regression provides the strongest overall performance for this dataset. Therefore, Linear Regression was selected as the final model.

8. Model Interpretation

The model comparison demonstrates why evaluating a machine learning project with multiple metrics is important. If only MAE were considered, Random Forest would appear to be the best choice. However, Random Forest produced a considerably higher RMSE and a lower R^2 Score than Linear Regression.

The Linear Regression result suggests that the available features contain useful overall linear relationships with house price. Its R^2 Score of 0.6369 indicates that a substantial portion of the variation in the target is captured by the model, although the remaining variation shows that house prices are also affected by factors not fully represented by the selected features or by relationships that are more complex than a simple linear model.

9. Conclusion

This project completed a full machine learning workflow for house price prediction. The dataset was inspected and cleaned, invalid price records were removed, and unnecessary prediction columns were excluded. Exploratory data analysis was then performed using six meaningful visualizations to understand price distributions and relationships among important property characteristics.

Feature preprocessing was applied to numerical and categorical variables, followed by an 80:20 train-test split. Linear Regression, Decision Tree Regressor, and Random Forest Regressor were trained and evaluated using MAE, MSE, RMSE, and R^2 Score.

Random Forest achieved the lowest MAE, but Linear Regression achieved the highest R^2 Score and the lowest MSE and RMSE. Based on the overall combination of evaluation results, **Linear Regression was selected as the final model**.

The project demonstrates how data cleaning, visualization, preprocessing, and model evaluation can be combined to develop and assess a practical house price prediction system.